08

Sets and classification

Part IV · Shape, structure and measure

CHAPTER 8 · OPENER

Sets and classification

Sets organise membership and overlap; clear definitions and representations prevent double counting and make survey claims testable.

ENGAGEMENT

When two groups overlap, what can we know about the missing intersection, and how does that change a decision?

Learning goals

- Use set notation, elements and subsets.
- Represent union, intersection and complement with Venn diagrams.
- Classify number families and place values on the real line.
- Interpret survey counts and justify decisions with stated denominators.

MYP lens

Key concept: Form

Related concepts: Representation, Systems

Global context: Fairness and development

ATL focus. In order for a student to organise information, students must define categories, represent overlap, check totals and explain how a definition changes a conclusion.

Language bank: set · element · subset · empty set · universal set · union · intersection · complement · Venn diagram · natural · whole · integer · rational · irrational · real

CHAPTER 8 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	List the even numbers from 1 to 8.	READY
R2	Calculate 17−6 and 30−25.	READY
R3	Find 2^3.	READY
R4	Write −3, 0, 2 in order.	READY
R5	What is 6/80 as a percentage?	READY

Chapter map

1. **8.1 Sets, elements and subsets** — Define a set clearly, list each element once and distinguish “element of” from “subset of”.
2. **8.2 Venn diagrams and set operations** — Place the overlap first, then the “only” regions, then the outside region.
3. **8.3 Number sets and the real line** — Classify a number by the sets it belongs to, not by the way it is written.
4. **8.4 Sets in surveys and decisions** — Define the universal set and categories before counting; state whether “or” includes overlap.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 8 · MATHEMATICAL DEVELOPMENT

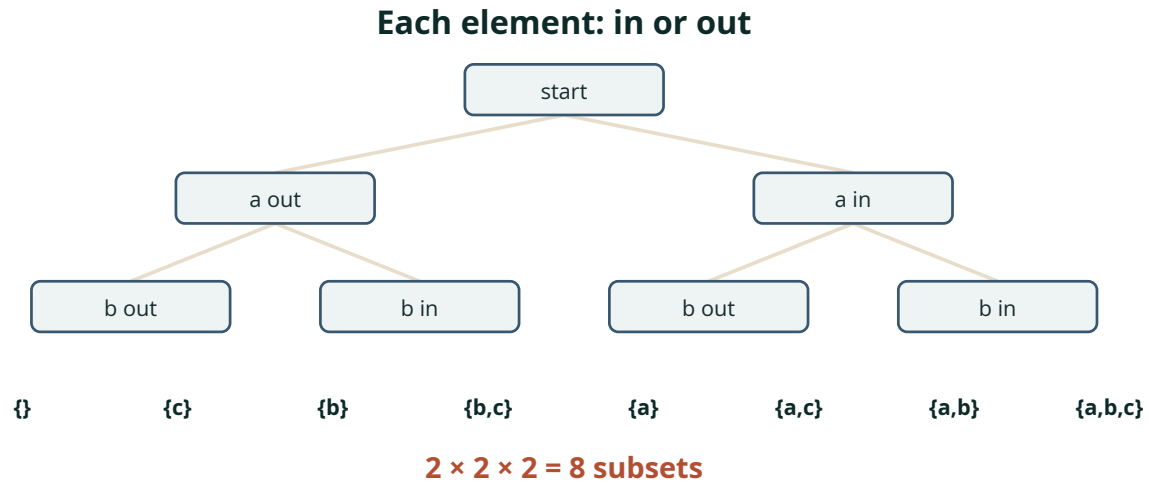
8.1 Sets, elements and subsets

BIG IDEA

Define a set clearly, list each element once and distinguish “element of” from “subset of”.

A set is a well-defined collection. $x \in A$ means x is an element of A; $B \subseteq A$ means every element of B is also in A. Order and repeated listings do not change a set.

The empty set $\emptyset$ has no elements and is a subset of every set. A set with n distinct elements has 2^n subsets, because each element is either included or excluded.



Each element has two choices: in or out. Three elements give $2^3 = 8$ subsets.

WORKED EXAMPLE · SUBSETS OF {A,B,C}

Choose included or excluded for each element.

1. 2×2×2 = 8 subsets.
2. ∅, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c}.

8 subsets.

CHAPTER 8 · PRACTICE

8.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

$\in$ relates an element to a set; $\subseteq$ relates one set to another. $\{a\} \subseteq \{a,b\}$, but $a \in \{a,b\}$.

GUIDED PRACTICE · FIRST STEPS

8.1.1 List the elements of A = {even numbers from 1 to 10}. FLUENCY

8.1.2 How many subsets has a 3-element set? A 4-element set? FLUENCY

8.1.3 Are {1,2,2,3} and {3,2,1} the same set? Explain. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.1.4 For $A=\{a,b\}$, decide: $a \in A$, $\{a\} \subseteq A$, $\emptyset \subseteq A$, $\{a\} \in A$. REASONING

8.1.5 A club chooses any subset of 5 activities, including none. How many choices? APPLICATION

8.1.6 How many subsets of {1,2,3,4,5} contain 1? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · MATHEMATICAL DEVELOPMENT

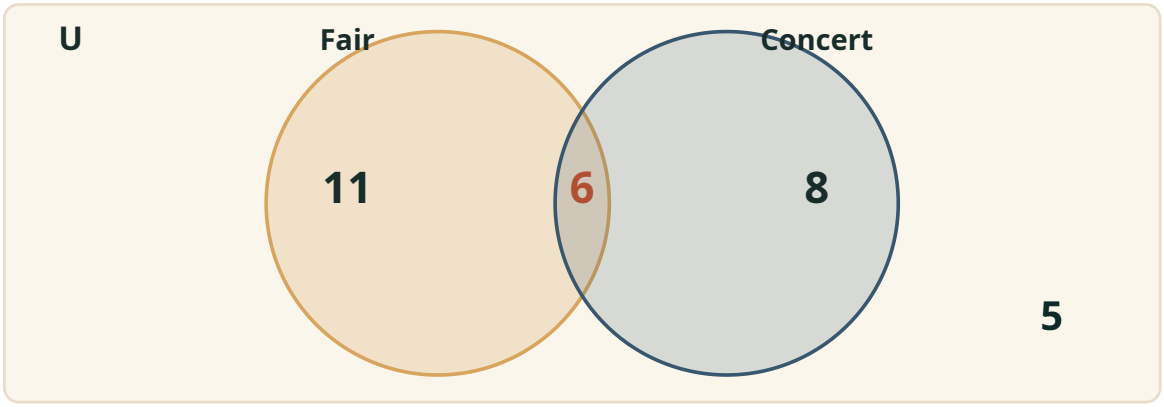
8.2 Venn diagrams and set operations

BIG IDEA

Place the overlap first, then the “only” regions, then the outside region.

Union $A \cup B$ contains elements in A or B or both. Intersection $A \cap B$ contains elements in both. Complement A' contains elements in the universal set ξ that are not in A.

For finite sets, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Subtract the overlap once because it was counted in both totals.



Total = 11 + 6 + 8 + 5 = 30

Overlap first: 11 + 6 + 8 + 5 = 30.

WORKED EXAMPLE · COMPLETE A VENN DIAGRAM

ξ has 30 students; 17 help at a fair, 14 at a concert, 6 at both.

1. Both: 6. Fair only: 17−6=11. Concert only: 14−6=8.
2. At least one: 11+6+8=25. Neither: 30−25=5.

Regions 11, 6, 8, 5.

CHAPTER 8 · PRACTICE

8.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

"A or B" in mathematics includes both unless the question says "exactly one".

GUIDED PRACTICE · FIRST STEPS

8.2.1 $A=\{1,2,3,4\}$, $B=\{3,4,5\}$. Find $A\cup B$ and $A\cap B$. FLUENCY

8.2.2 In a 30-student group, 17 like F, 14 like C and 6 like both. Find all four Venn regions. FLUENCY

8.2.3 Why is $n(A\cup B)$ not always $n(A)+n(B)$? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.2.4 Of 40 people, 22 take tea, 18 take coffee and 7 take both. How many take neither? APPLICATION

8.2.5 If $A\subseteq B$, what are $A\cap B$ and $A\cup B$? REASONING

8.2.6 $n(A)=12$, $n(B)=9$ and $n(A\cup B)=17$. Find $n(A\cap B)$. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · MATHEMATICAL DEVELOPMENT

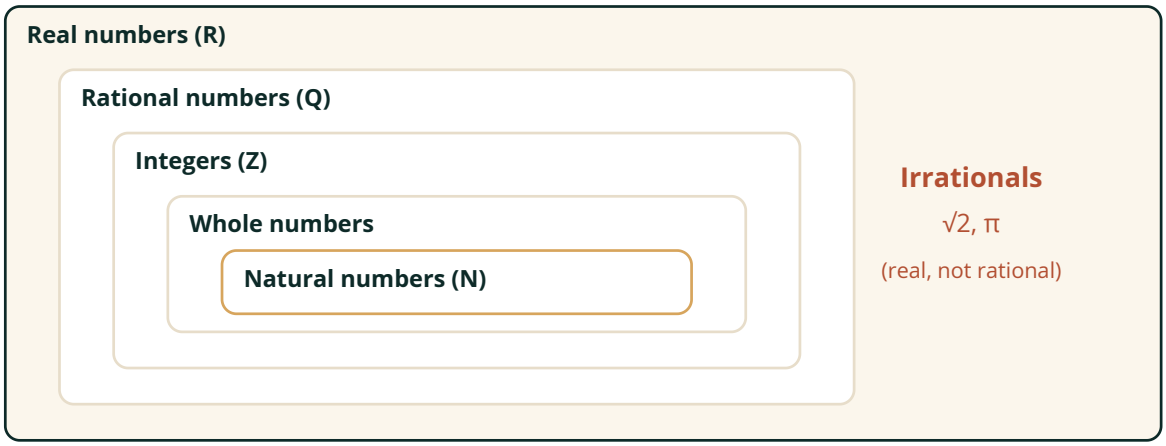
8.3 Number sets and the real line

BIG IDEA

Classify a number by the sets it belongs to, not by the way it is written.

Natural numbers $\mathbb{N}$ are counting numbers (here 1,2,3,...); whole numbers add 0; integers $\mathbb{Z}$ add negatives; rationals $\mathbb{Q}$ are ratios of integers with nonzero denominator. Real numbers $\mathbb{R}$ include rationals and irrationals.

Thus natural numbers are a subset of whole numbers, which are a subset of integers, which are a subset of rationals, which are a subset of reals. A terminating or repeating decimal is rational. Non-repeating, non-terminating decimals such as $\sqrt{2}$ and π are irrational.



Natural numbers sit inside whole numbers, integers, rationals and reals. Irrationals are real but not rational.

WORKED EXAMPLE · CLASSIFY −3, 0, 0.75 AND √2

Use the most specific familiar set, then note wider membership.

1. −3 is an integer and rational.
2. 0 is whole, integer and rational.
3. 0.75 = 3/4 is rational.
4. $\sqrt{2}$ is irrational and real.

All four are real; only $\sqrt{2}$ is irrational.

CHAPTER 8 · PRACTICE

8.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A number can belong to several sets. “Rational” does not mean “fraction-looking”; 0.5 and 1/2 are both rational.

GUIDED PRACTICE · FIRST STEPS

8.3.1 Classify −4, 0, 5 and 1/2 as natural, whole, integer, rational or real (give the most specific listed). FLUENCY

8.3.2 Which are irrational: √2, π, 0.25, 1/3? FLUENCY

8.3.3 Is every integer rational? Is every rational an integer? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.3.4 Place −2, 0, 1/2 and 2 on a number line in order. APPLICATION

8.3.5 Explain why 0.333… is rational. REASONING

8.3.6 Find a rational and an irrational number between 1 and 2. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · MATHEMATICAL DEVELOPMENT

8.4 Sets in surveys and decisions

BIG IDEA

Define the universal set and categories before counting; state whether “or” includes overlap.

A survey count is only meaningful with its universal set. “Pool users” may include people who also use the gym; “pool only” excludes them.

Use a Venn diagram or table to organise counts, check that regions sum to the total, and translate the result into a decision with a stated assumption.

WORKED EXAMPLE · A CENTRE AUDIT

80 members: 44 pool, 37 gym, 18 both.

1. Pool only 26; gym only 19; both 18; neither 17.
2. At least one: 63, which is 78.75% of 80.

Regions 26, 18, 19, 17; 78.75% use at least one.

CHAPTER 8 · PRACTICE

8.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A percentage needs its denominator. 37/80 and 19/80 answer different questions about gym use.

GUIDED PRACTICE · FIRST STEPS

8.4.1 80 members: 44 pool, 37 gym, 18 both. Find the four regions. APPLICATION

8.4.2 What percentage use at least one facility? APPLICATION

8.4.3 Is “more than a third use the gym” true? Discuss both meanings. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

8.4.4 Pool-only pass $4, gym-only $3, combined $6. Find income if each member buys the relevant pass. APPLICATION

8.4.5 Why should a survey report state the universal set? REASONING

8.4.6 50 people: 34 like vanilla, 29 chocolate, 7 neither. Find both and vanilla only. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 8 · EXPLORATION

How much overlap is possible?

A committee has 30 students. 17 help at a fair (F), 14 at a concert (C). First use a stated overlap of 6; then treat the overlap x as unknown.

> **INVESTIGATION**
> Criterion B. Use inequalities to find the full range of a missing intersection, then justify both endpoints.

QUANTITY	EXPRESSION OR VALUE
n(F)	17
n(C)	14
n(F∩C)	x
n(F∪C)	
Neither	

EXPLORATION · HOW MUCH OVERLAP IS POSSIBLE?

Criterion B. Use inequalities to find the full range of a missing intersection, then justify both endpoints.

B1 With overlap 6, find all four regions and the number helping at least one event. INVESTIGATION

B2 With unknown overlap x, write n(F∪C) and the neither count. INVESTIGATION

B3 Find the smallest possible x and explain why x=0 fails. INVESTIGATION

B4 Find the largest possible x and describe that diagram. INVESTIGATION

B5 The committee needs 20 helpers for a combined event. With overlap 6, are there enough? Could the answer change across the possible range? CHALLENGE

CHAPTER 8 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	How many subsets has {a,b,c,d}?	CHECK
C2	A={1,2,3}, B={2,3,4}. Find A∩B.	CHECK
C3	n(A)=10, n(B)=8, overlap 3. Find n(A∪B).	CHECK
C4	Classify 0.6 as rational or irrational.	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 8 · APPLICATION

The centre audit

80 members: 44 use the pool, 37 use the gym, 18 use both. Passes: pool $4, gym $3, combined $6.

> **REAL-LIFE APPLICATION**
> Criterion D. Simulated membership data. State the region and denominator behind every claim.

REGION	COUNT
Pool only	
Both	
Gym only	
Neither	

APPLICATION · THE CENTRE AUDIT

Criterion D. Simulated membership data. State the region and denominator behind every claim.

D1	Complete the four regions and check the total. APPLICATION
D2	Find pass income and the percentage using at least one facility. APPLICATION
D3	Test “more than a third use the gym” under two meanings. REASONING
D4	A combined pass costs $6 instead of $4+$3. Give one member argument, one centre argument and a recommendation. CHALLENGE

CHAPTER 8 · ASSESSMENT PRACTICE

The classification poster

MYP ASSESSMENT PRACTICE

Criteria A and C. Create a poster that distinguishes membership, subsets and number families, with one checked example.

Your poster should prevent the common confusion between $\in$ and $\subseteq$ and between rational and irrational.

ASSESSMENT PRACTICE · THE CLASSIFICATION POSTER

Criteria A and C. Create a poster that distinguishes membership, subsets and number families, with one checked example.

S1	Give one correct $\in$ statement and one correct $\subseteq$ statement using $\{1,2,3\}$.	ASSESSMENT
S2	Classify -2, 0, $3/4$ and $\sqrt{2}$, and state one nesting fact.	ASSESSMENT
S3	Write a check question that exposes double counting, with its answer.	ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 8 · MIXED PRACTICE

Mixed practice 8

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 8

Connect ideas. Give units and reasons where requested.

M1	How many subsets has a 5-element set?	FLUENCY
M2	A={2,4,6}, B={4,6,8}. Find union and intersection.	FLUENCY
M3	In 50 people, 28 like A, 22 like B, 9 both. Find neither.	APPLICATION
M4	Is 0.121212... rational? Explain.	REASONING
M5	Classify √9 and √10.	FLUENCY
M6	If A⊆B, n(A)=7 and n(B)=12, find n(A∩B) and n(A∪B).	REASONING
M7	60 people: 35 use X, 28 use Y, 10 neither. Find both.	APPLICATION
M8	n(A)=20, n(B)=15, universal 30. Find the possible overlap range.	CHALLENGE
M9	Explain why {1,2} and {1,1,2,2} are equal.	REASONING
M10	A set has 6 elements. How many subsets contain a particular element?	APPLICATION

> **FINAL REFLECTION**
>
> Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 8 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **2, 4, 6, 8.**
Apply the rule.

R2 **11 and 5.**
Subtract the overlap and then the union.

R3 **8.**
2×2×2.

R4 **−3, 0, 2.**
Negatives lie left of zero.

R5 **7.5%.**
6÷80×100.

CHECKPOINT

C1 **16.**
2^4.

C2 **{2,3}.**
Keep shared elements.

C3 **15.**
10+8−3.

C4 **Rational: 3/5.**
It is a ratio of integers.

CHAPTER 8 · ANSWERS AND REASONING

Answers · 8.1 and 8.2

PRACTICE 8.1 · SETS AND SUBSETS

8.1.1 **{2, 4, 6, 8, 10}.**
Apply the membership rule once to each candidate.

8.1.2 **8 and 16.**
Use 2^n.

8.1.3 **Yes. Repetition and order do not matter; both contain exactly 1, 2 and 3.**
A set records membership, not listing order.

8.1.4 **True, true, true, false (unless {a} itself were listed as an element).**
Distinguish elements from subsets.

8.1.5 **32.**
2^5.

8.1.6 **16.**
Fix 1 as included; choose freely among the other four.

PRACTICE 8.2 · VENN OPERATIONS

8.2.1 **{1,2,3,4,5} and {3,4}.**
Union collects all; intersection keeps shared elements.

8.2.2 **F only 11; both 6; C only 8; neither 5.**
Subtract the overlap from each total; subtract the union from 30.

8.2.3 **The intersection is counted twice, so subtract it once.**
Use inclusion–exclusion.

8.2.4 **7.**
Union 22+18−7=33; 40−33.

8.2.5 **A and B.**
Every element of A is already in B.

8.2.6 **4.**
12+9−17.

CHAPTER 8 · ANSWERS AND REASONING

Answers · 8.3 and 8.4

PRACTICE 8.3 · NUMBER FAMILIES

8.3.1 **−4 integer; 0 whole; 5 natural; 1/2 rational. All are real.**
Use the nesting of number sets.

8.3.2 **√2 and π.**
0.25=1/4 and 1/3 are ratios of integers.

8.3.3 **Yes; no. For example −3=−3/1 is rational, but 1/2 is not an integer.**
Integers are a subset of rationals.

8.3.4 **−2 < 0 < 1/2 < 2.**
Negative values lie left of zero.

8.3.5 **It equals 1/3, a ratio of integers.**
Repeating decimals are rational.

8.3.6 **Sample: 3/2 and √2.**
1.5 lies between; √2≈1.414 also lies between.

PRACTICE 8.4 · SURVEY COUNTS

8.4.1 **26, 18, 19, 17.**
Subtract overlap, then subtract union from 80.

8.4.2 **78.75%.**
63/80×100.

8.4.3 **Yes for all gym users: 37/80=46.25%. No for gym only: 19/80=23.75%.**
The category definition changes the claim.

8.4.4 **$26×4 + 19×3 + 18×6 = $269.**
Use the exactly-one and both regions.

8.4.5 **The same count can mean different proportions in different populations.**
Percentages and “neither” depend on the total and the denominator.

8.4.6 **Both 20; vanilla only 14.**
Union 43; overlap 34+29−43.

CHAPTER 8 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · HOW MUCH OVERLAP IS POSSIBLE?

B1 **11, 6, 8, 5; at least one 25.**
Subtract the overlap and use the total 30.

B2 **31−x and x−1.**
17+14−x; 30−(31−x).

B3 **x=1. If x=0, the union would be 31, more than the 30 students.**
The union cannot exceed the universal set.

B4 **x=14. All concert helpers also help at the fair; C is a subset of F.**
The overlap cannot exceed either set total.

B5 **Yes, 25 helpers. Across x=1 to 14, the union ranges from 30 down to 17, so for x≥12 there would be fewer than 20.**
Union = 31−x; solve 31−x≥20.

CHAPTER 8 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE CENTRE AUDIT

D1 **26, 18, 19, 17; sum 80.**
Pool only 44−18; gym only 37−18; neither 80−63.

D2 **$269; 78.75%.**
26×4+19×3+18×6; 63/80×100.

D3 **All gym users: true (46.25%). Gym only: false (23.75%).**
State whether the overlap is included.

D4 **Sample: members save $1; the centre may receive less per dual user but gain simplicity and uptake. Recommend tracking visits and uptake before changing price.**
A fairness judgment needs a stated criterion and data.

ASSESSMENT PRACTICE · THE CLASSIFICATION POSTER

S1 **Sample: $2 \in \{1,2,3\}$; $\{1,2\} \subseteq \{1,2,3\}$.**
The first relates an element; the second relates sets.

S2 **Integer, whole, rational, irrational; all real.** $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
Use the most specific listed family and wider membership.

S3 **Sample: n(A)=10, n(B)=8, overlap 3. Union? 15, not 18.**
Subtract the overlap once.

CHAPTER 8 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 8

M1 **32.**
2^5.

M2 **{2,4,6,8}; {4,6}.**
Collect all; keep shared.

M3 **9.**
50−(28+22−9).

M4 **Yes; repeating decimals are ratios of integers (12/99 = 4/33).**
A repeating block can be converted to a fraction.

M5 **√9=3 is natural/rational; √10 is irrational.**
Simplify perfect squares first.

M6 **7 and 12.**
A is entirely inside B.

M7 **13.**
Union 50; overlap 35+28−50.

M8 **5 to 15.**
max(0,20+15−30) ≤ x ≤ min(20,15).

M9 **Sets ignore repetition; both have exactly elements 1 and 2.**
Membership defines a set.

M10 **32.**
Fix it in; choose among the other five: 2^5.